

Virtual Human in Symmetrical Reality: A Mobile System for Human-AI Coexistence

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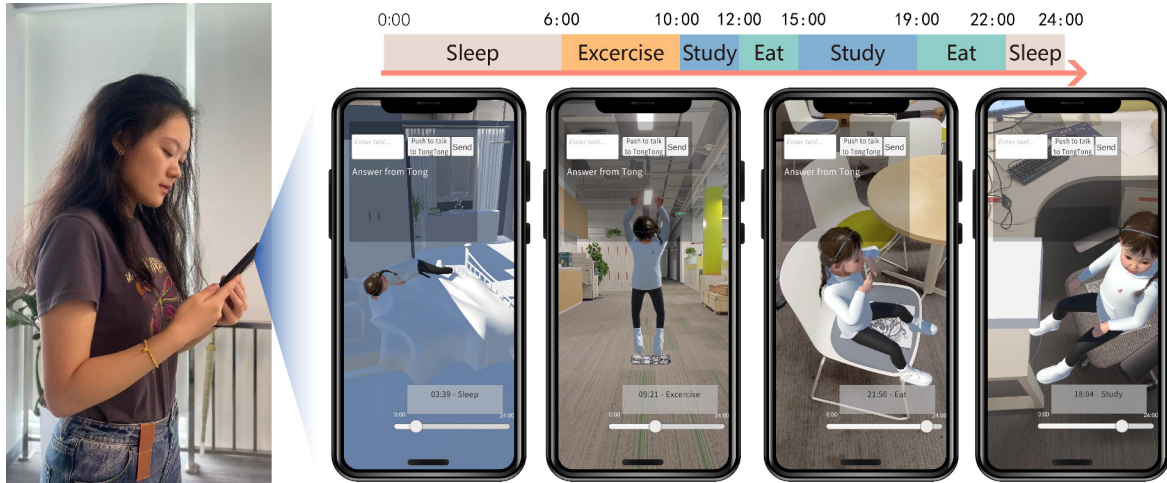


Figure 1: The system presents a mobile mixed-reality framework where an AR companion agent accompanies the user across a full day. Using timeline-based scheduling, it instantiates, animates, and localizes the agent in the physical environment to provide continuous, context-aware presence and companionship.

ABSTRACT

The advancement of artificial intelligence has intensified the challenge of creating intelligent virtual human capable of sustaining long-term companionship. This paper presents a general framework for a virtual companion system based on augmented reality (AR), which prioritizes four key dimensions: situational coherence, immersive interactivity, conversational naturalness, and continuity of presence. We instantiate this framework in a functional AR prototype for handheld mobile devices and develop a comparable non-AR system for a controlled comparison. Through comprehensive user studies, we evaluate the system's effectiveness across the proposed dimensions. Our findings offer important insights and a valuable reference for the future development of companion virtual human.

Index Terms: Interactive systems and tools, User studies.

1 INTRODUCTION

Advances in AI have accelerated the emergence of symmetrical reality, reshaping how people interact with AI-driven virtual humans across physical and virtual spaces. As a practical instantiation, augmented reality (AR) brings virtual humans into everyday environments and highlights their potential as companions: unlike 2D screens or fully immersive VR, AR embeds an agent into the real world so users can perceive it as situated and co-present. This paper asks: *How can an AR virtual human be persistently perceived*

as a credible, interactive, and natural friend in real environments?

We view the challenge as spanning four coupled layers—space, interaction, communication, and time—where misalignment in any layer can undermine companionship.

From a user-experience perspective, AR companionship should be evaluated along four dimensions. *Situational coherence* reflects whether spatial registration and semantically appropriate behavior make the agent feel co-located. *Immersive interactivity* depends on a responsive physical-virtual loop that turns user or environmental changes into interpretable agent reactions. *Conversational naturalness* requires coordination between language and nonverbal cues such as gaze and posture to match social norms. *Continuity of presence* depends on smooth temporal organization of entrances, exits, and state transitions along a single time axis.

Accordingly, we test four hypotheses in this work: **H1** situation-oriented alignment and semantic consistency improve situational coherence; **H2** a bidirectional physical-virtual loop strengthens immersion and interactive experience; **H3** coupling dialogue with gaze/posture enhances conversational naturalness and perceived companionship; and **H4** smooth entrances/exits and path animations maintain temporal continuity and realism of presence.

2 SYSTEM DESIGN

2.1 AR System

Guided by H1–H4, we implement an AR companion agent with four modules (A–D) that support daily scheduling, object-aware interaction, conversational feedback, and motion control. The agent follows four routine states (*study, meal, sleep, exercise*) and uses a state identifier to ensure smooth temporal transitions. Physical markers are mapped to semantic locations so the agent's behavior remains aligned with the real space, yielding a coherent, end-to-end AR experience.

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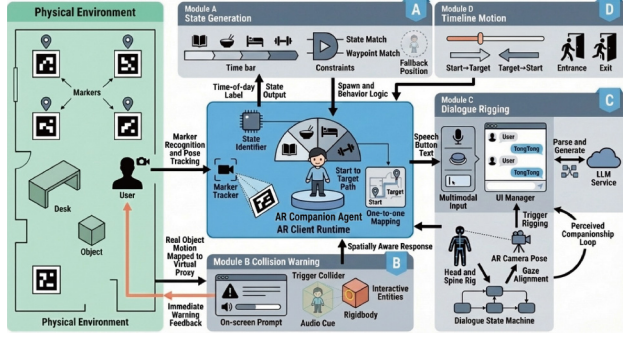


Figure 2: The system has four modules: (A) state generation for routine states and transitions, (B) collision warning for realistic physical feedback, (C) dialogue rigging to synchronize responses with embodied actions, and (D) timeline motion to align behavior with time-of-day changes.

The state generation module maps time-of-day to one of four routine states and selects a matched waypoint pair (*Start*, *Target*) from tracked markers. When both state and waypoint constraints are satisfied, the AR companion agent enters from a predefined fallback position and moves along the *Start*→*Target* path; otherwise, it stays at the fallback position to avoid abrupt appearances. This design enables stable state switching while preserving tight spatial coupling to the physical environment.

The collision warning module binds tagged real objects to virtual proxies via markers and streams their poses into the virtual scene. When the AR companion agent contacts a tagged object, the system immediately presents a visual warning (optionally with audio), then fades and resets after separation. This loop makes physical object manipulation directly influence perceived safety and interactivity. By grounding virtual responses in real-world object manipulation, this feedback loop addresses a common limitation of AR agents that are visually present but physically inconsequential, thereby enhancing perceived safety and embodied interactivity.

The dialogue rigging module supports multimodal user input (speech and text) and sends it through a communication layer to an LLM service for parsing and response generation. Returned utterances are integrated into a multi-turn dialogue state machine and synchronized with animation rigging so head, spine, and gaze align with the user’s viewpoint during responses. This coupling produces spatially consistent conversation and strengthens the agent’s perceived presence. By coupling linguistic output with embodied cues, the system avoids disembodied, voice-only interaction and produces spatially consistent conversation that aligns verbal responses with social and spatial norms, thereby strengthening the agent’s perceived presence.

The timeline motion module controls entrance and exit by matching the current time-state to waypoint pairs. When a valid (*Start*, *Target*) match exists, the AR companion agent follows *Start*→*Target* with an entrance animation; when the time-state changes or matching is lost, it follows *Target*→*Start* with an exit animation. This mechanism preserves coherent spatial logic across daily transitions and improves motion realism. This symmetric motion logic preserves spatial coherence across daily transitions and prevents abrupt appearance or disappearance, contributing to more realistic motion behavior and a continuous sense of presence over time.

2.2 Non-AR System as Comparison

For evaluation, we implement a non-AR baseline that preserves the same animations, dialogue content, task logic, time-state design, and triggers, but removes AR spatial presentation. All content is

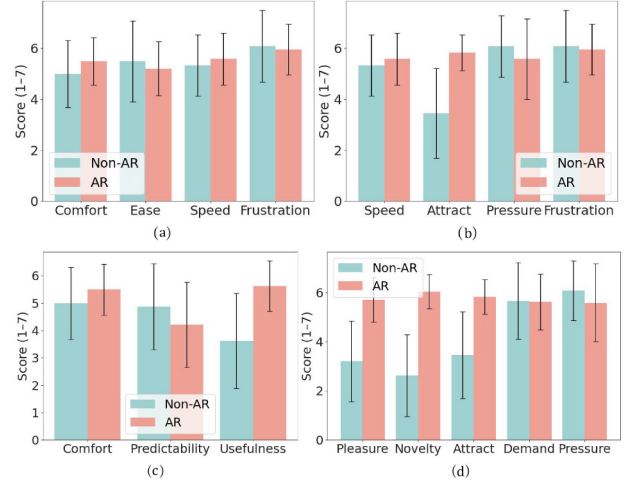


Figure 3: Bar plots summarize user study results for H1–H4. (a–b) Situational Coherence (H1) and Immersive Interactivity (H2): the AR system improves Safety, Functional Relevance, Enjoyment, Novelty, and Attractiveness without increasing Demand or Pressure. (c–d) Conversational Naturalness (H3) and Continuity of Presence (H4): the AR system enhances Comfort, Ease of Use, Speed, and Attractiveness while reducing Pressure and Frustration, indicating improved interaction quality without added cognitive or physical strain.

delivered on a standard 2D display, enabling observed differences to be attributed to AR spatial interaction rather than non-spatial confounds.

3 EXPERIMENT

We recruited 24 participants (ages 25–35) with normal or corrected vision/hearing and no prior exposure to the systems. Using a within-subjects, counterbalanced design, each participant used both an AR system and a non-AR 3D baseline to complete state switching and voice dialogue tasks, with the AR condition additionally requiring marker-based state triggering, collision-warning tests with real objects, and timeline-driven transition motion. The Unity app ran on a Xiaomi 13 Ultra Pro Max and used Vuforia for marker tracking; the study took place in an $\sim 200\text{m}^2$ indoor space mapped to four daily states. We measured outcomes with UEQ-S, NASA-TLX, and an 8-item custom questionnaire aligned with H1–H4, and each 10–20 minute session ended with a brief interview.

4 CONCLUSION AND FUTURE WORK

We propose a four-dimensional AR companion evaluation framework and show AR outperforms a non-AR baseline in perceived safety, relevance, and immersion, enabling reproducible cross-media assessment.

Our modular prototype supports applications such as education and rehabilitation. Priorities include predictive cues, low-latency physical–virtual feedback, and synchronized language–posture co-ordination, with future work on multi-user/longitudinal settings, objective sensing, and markerless tracking via semantic SLAM.

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